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# =====  
# Real-Time Surveillance System using Computer Vision  
# Google Colab  
# =====  
  
import cv2  
import matplotlib.pyplot as plt  
from google.colab import files  
  
# Upload video  
uploaded = files.upload()  
  
video_name = list(uploaded.keys())[0]  
  
# Open video  
cap = cv2.VideoCapture(video_name)  
  
# Background subtractor for motion detection  
fgbg = cv2.createBackgroundSubtractorMOG2()  
  
frame_count = 0  
  
while True:  
    ret, frame = cap.read()  
  
    if not ret or frame_count >= 5:    # Show first 5 processed frames  
        break  
  
    frame = cv2.resize(frame, (640, 480))  
  
    # Detect moving objects  
    mask = fgbg.apply(frame)  
  
    # Find contours  
    contours, _ = cv2.findContours(mask,  
                                    cv2.RETR_EXTERNAL,  
                                    cv2.CHAIN_APPROX_SIMPLE)
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for cnt in contours:
    if cv2.contourArea(cnt) > 500:
        x, y, w, h = cv2.boundingRect(cnt)
        cv2.rectangle(frame,
                        (x, y),
                        (x+w, y+h),
                        (0,255,0),
                        2)

# Convert to RGB for display
rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(6,4))
plt.imshow(rgb)
plt.title(f"Processed Frame {frame_count+1}")
plt.axis('off')
plt.show()

frame_count += 1

cap.release()

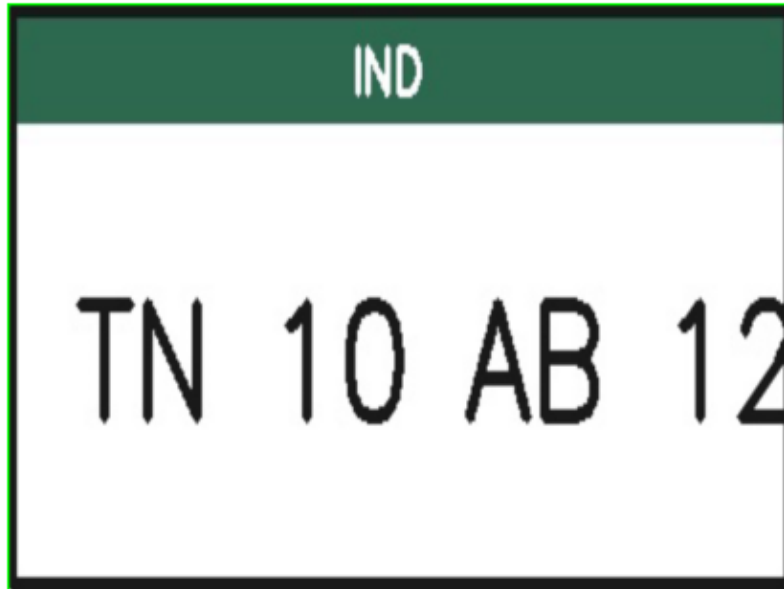
print("\nSurveillance Analysis")
print("-----")
print("Level of Computer Vision : High-Level Vision")
print("Reason:")
print("- Captures video continuously.")
print("- Detects moving objects.")
print("- Interprets the scene automatically.")
print("- Suitable for real-time surveillance applications.")
```

number_plate.jpg

number_plate.jpg(image/jpeg) - 14891 bytes, last modified: 7/13/2026 - 100% done

Saving number_plate.jpg to number_plate.jpg

Processed Frame 1



Surveillance Analysis

Level of Computer Vision : High-Level Vision

Reason:

- Captures video continuously.
- Detects moving objects.
- Interprets the scene automatically.
- Suitable for real-time surveillance applications.

